Fundamental Trigonometric Identities

Using the Pythagorean identities

1 Given that $\sin\theta = \frac{3}{5}$ and θ lies in Quadrant II, find:

a $\cos\theta$ **b** $\tan\theta$ **c** $\sec\theta$ **d** $\csc\theta$

2 Given that $\tan\theta = 2$ and θ lies in Quadrant III, find $\sin\theta$ and $\cos\theta$ in exact simplified form.

Simplifying and verifying

3 Simplify each expression to a single trigonometric function:

a $\frac{1-\cos^2 x}{\sin x}$

b $\sec x\cot x$

c $\tan x\csc x$

4 Verify the identity $(1+\tan^2 x)\cos^2 x = 1$.

5 Verify the identity $\frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$.